

PROBABILISTIC SYSTEMS ANALYSIS AND APPLIED PROBABILITY

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LECTURE 1: PROBABILITY MODELS AND AXIOMS

Probability gives us a way to calculate with uncertainty. A model describes what can happen and assigns probabilities to events. We first define these two parts, then use the probability axioms to work through discrete and continuous examples.

1.1 The Sample Space

Definition 1.1 [SAMPLE SPACE]. The sample space Ω is the set, or list, of possible outcomes of an experiment. It lists what can happen in one run of the experiment.

The list of outcomes is not arbitrary. It must be **MUTUALLY EXCLUSIVE**, so that at most one outcome occurs, and **COLLECTIVELY EXHAUSTIVE**, so that at least one of them does. Together these say that when the experiment is run, exactly one element of Ω is realized.

Choose enough detail to describe the events you care about, but leave out irrelevant information. For an ordinary coin-toss problem, recording heads or tails is enough; the coin's temperature does not help.

Example: Roll a tetrahedral (four-sided) die twice. Taking the outcome to be the ordered pair of face values,

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\},$$

which has 16 elements. The same experiment admits a sequential description as a tree ([figure 1](#)): a first branching into the four values of x , and below each of those a second branching into the four values of y . The tree and the 4×4 grid describe the same sample space; the tree merely emphasizes that the rolls happen in order.

A sample space need not be finite, or even countable.

Example: Pick a point from the unit square. Here

$$\Omega = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\},$$

an uncountably infinite set of outcomes.

1.2 Events and the Axioms

Definition 1.2 [EVENT]. An event is a subset of the sample space. It collects the outcomes that satisfy a condition we care about.

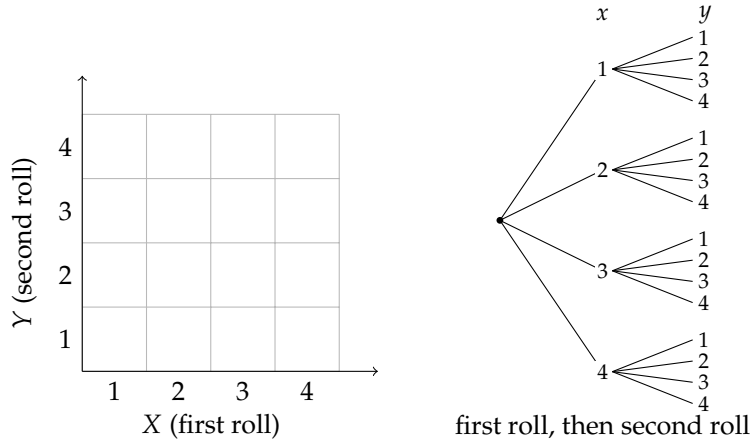


Figure 1: Two descriptions of the sample space for two rolls of a tetrahedral die. Left: the 4×4 grid of ordered pairs (x, y) . Right: the sequential tree, branching first on x and then on y ; the leaf reached through branches x and y is the outcome (x, y) , so the 16 leaves are the same 16 outcomes.

Probabilities are assigned to events. For a uniformly chosen point in a square, each exact point has probability zero, but a region such as the left half has positive probability. We therefore need probabilities of sets of outcomes, not just individual outcomes.

Definition 1.3 [PROBABILITY LAW]. A probability law assigns to each event A a number $\mathbb{P}(A)$, the probability of A , subject to three axioms:

1. **NONNEGATIVITY:** $\mathbb{P}(A) \geq 0$ for every event A .
2. **NORMALIZATION:** $\mathbb{P}(\Omega) = 1$.
3. **ADDITIVITY:** if $A \cap B = \emptyset$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Everything else is derived. Applying additivity repeatedly to a finite collection of distinct outcomes $s_1, \dots, s_k$ gives

$$\mathbb{P}(\{s_1, s_2, \dots, s_k\}) = \mathbb{P}(\{s_1\}) + \dots + \mathbb{P}(\{s_k\}) = \mathbb{P}(s_1) + \dots + \mathbb{P}(s_k),$$

writing $\mathbb{P}(s)$ for $\mathbb{P}(\{s\})$ as is customary. So on a finite sample space a probability law is completely determined by the probabilities of the individual outcomes.

Note. Two details need care. Finite additivity does not cover an infinite list of events; we strengthen it below. Also, in continuous models, probabilities are defined on a chosen family of measurable sets. Intervals, rectangles, and countable combinations of them are included; arbitrary subsets need not be.

1.3 A Probability Law on a Finite Sample Space

Return to the two rolls of a tetrahedral die, and let each of the 16 outcomes have probability $1/16$. Since the outcome probabilities determine the law, every event can now be computed by counting cells, as in [figure 2](#).

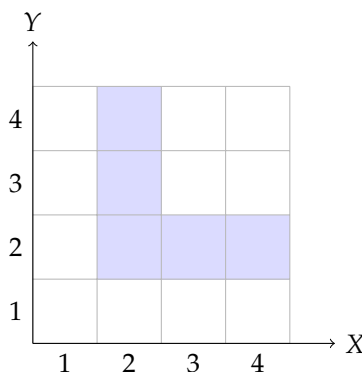


Figure 2: The sample space for two rolls of a tetrahedral die, with X the first roll and Y the second. The five shaded cells are the event $\{\min(X, Y) = 2\}$.

- $\mathbb{P}((X, Y) \text{ is } (1, 1) \text{ or } (1, 2)) = \frac{2}{16} = \frac{1}{8}$, two cells.
- $\mathbb{P}(X = 1) = \frac{4}{16} = \frac{1}{4}$, the column of four cells above $X = 1$.
- $\mathbb{P}(X + Y \text{ is odd}) = \frac{8}{16} = \frac{1}{2}$: the sum is odd exactly when one roll is odd and the other even, which happens in $2 \cdot 2 + 2 \cdot 2 = 8$ cells.
- $\mathbb{P}(\min(X, Y) = 2) = \frac{5}{16}$: the minimum is at least 2 in the $3 \times 3 = 9$ cells with both rolls in $\{2, 3, 4\}$ and at least 3 in the $2 \times 2 = 4$ cells with both in $\{3, 4\}$, leaving $9 - 4 = 5$.

1.4 The Discrete Uniform Law

The same counting rule works for any finite set of equally likely outcomes.

Theorem 1.4 [DISCRETE UNIFORM LAW]. *If Ω is finite and all outcomes are equally likely, then for every event A ,*

$$\mathbb{P}(A) = \frac{\text{number of elements of } A}{\text{total number of sample points}}.$$

Note. For equally likely outcomes, probability is a counting problem. A fair coin, a fair die, and a well-shuffled deck each use a uniform law on the relevant sample space.

1.5 The Continuous Uniform Law

Now take two “random” numbers in $[0, 1]$, meaning a point drawn from the unit square with no region favored over another of the same size. The uniform law here assigns each event its area:

$$\mathbb{P}(A) = \text{area of } A.$$

This satisfies the axioms: areas are nonnegative, the unit square has area 1, and disjoint regions have areas that add.

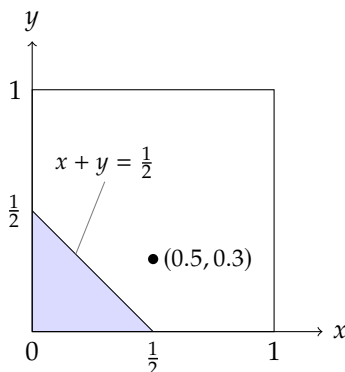


Figure 3: The continuous uniform law on the unit square. The shaded triangle $\{x + y \leq 1/2\}$ has probability equal to its area, $1/8$; the single point $(0.5, 0.3)$ has area, and hence probability, zero.

Example: $\mathbb{P}(X + Y \leq 1/2)$ is the area of the shaded triangle in [figure 3](#), with vertices $(0, 0)$, $(1/2, 0)$, and $(0, 1/2)$, namely $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$.

Example: $\mathbb{P}((X, Y) = (0.5, 0.3)) = 0$, since a single point has zero area.

Every exact point has probability zero, yet one point is selected. Probability zero therefore need not mean impossible. Probabilities of regions describe where the selected point is likely to lie.

1.6 Countable Additivity

The gap left by Axiom 3 appears as soon as the sample space is countably infinite.

Example: Let $\Omega = \{1, 2, \dots\}$ with $\mathbb{P}(n) = 2^{-n}$ for each n . What is the probability that the outcome is even?

The natural computation ([figure 4](#)) sums a geometric series:

$$\mathbb{P}(\{2, 4, 6, \dots\}) = \mathbb{P}(2) + \mathbb{P}(4) + \dots = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots = \frac{1/4}{1 - 1/4} = \frac{1}{3}.$$

But this breaks an infinite union into infinitely many pieces, and finite additivity does not justify that step. The axiom must be strengthened.

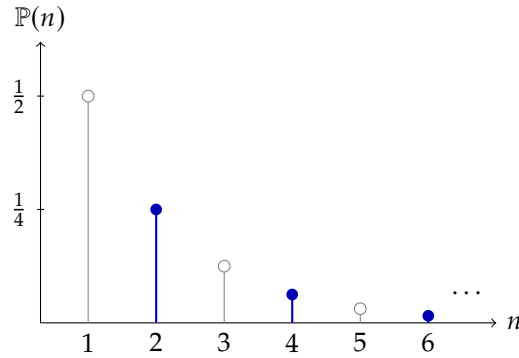


Figure 4: The law $\mathbb{P}(n) = 2^{-n}$ on $\{1, 2, \dots\}$. The filled stems mark the even outcomes, whose probabilities sum to $1/3$.

Definition 1.5 [COUNTABLE ADDITIVITY]. If $A_1, A_2, \dots$ is a sequence of pairwise disjoint events, then

$$\mathbb{P}(A_1 \cup A_2 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots .$$

Countable additivity replaces Axiom 3 and implies it, taking all but two of the events empty. Countable is also as far as one can go: additivity over an *uncountable* family is false, as the unit square already shows. There the whole square is the disjoint union of its individual points, each of probability zero, and no sum of zeros gives 1.

Source: MIT 6.041 Probabilistic Systems Analysis and Applied Probability, Fall 2010, Lecture 1 notes by John Tsitsiklis.